
DATA ANALYST

Complete 2-Month Training Roadmap

10 Core Modules · Structured Weekly Plan · Job-Ready Skills

Excel · SQL · Statistics · Python · Data Cleaning

EDA · Visualization · Streamlit · Power BI · Business Analytics · ML

Why This Course Exists — The Story Behind It

The Problem We Saw

Every year, hundreds of students graduate with degrees in commerce, science, or engineering — and when they sit across from a hiring manager at a data analyst role, they freeze. Not because they aren't smart. But because nobody taught them the right things in the right order. They learned Python in college but never touched real messy data. They know Excel but have never built a dashboard a manager would actually open. They've heard the word 'SQL' but never written a query against a real database. This course was built to fix that gap — completely.

Why We Start With Excel

You might wonder why a modern data course starts with Excel. Here's the truth: walk into any company in India — a startup, a bank, an e-commerce firm — and open their analyst's laptop. You will almost certainly find Excel. It is the universal language of business. Before we teach anyone to write code, we make sure they can work at the speed of business. Excel is that foundation.

Why Statistics Comes Early

A lot of courses skip statistics or dump it at the end as an afterthought. We disagree. If you don't understand what an average actually means, what a distribution looks like, or how to test whether a trend is real or just noise — you will make wrong decisions confidently. Statistics is the 'why behind the numbers.' We teach it early so everything else makes sense.

Why SQL Is Non-Negotiable

Data lives in databases. Whether a company uses MySQL, PostgreSQL, or Snowflake, one skill is constant: SQL. It's the language that unlocks the data. Every analyst job posting in the world asks for it. We teach SQL as a practical querying skill — not as theory — so students can pull, filter, join, and summarize data from day one of a job.

Why Python — Not Just 'Coding'

We don't teach Python for the sake of programming. We teach it because Pandas, NumPy, and visualization libraries let an analyst do in seconds what would take hours in Excel. Python is the power tool. But we frame every lesson around data problems, not syntax. The goal is always: solve a real data question faster and smarter.

Why Data Cleaning Gets Its Own Module

In the real world, 60–80% of an analyst's time is spent cleaning data before any analysis begins. Yet most courses treat cleaning as a footnote. Not here. We dedicate an entire module to it — missing values, outliers, duplicates, transformations — because if you can't clean data, you can't analyze it. Simple as that.

Why Visualization Includes Streamlit

Charts are only useful if they reach the right person. Matplotlib and Seaborn produce static charts for reports. Power BI builds enterprise dashboards. But Streamlit is the middle ground that modern analysts are increasingly expected to know — it lets you turn a Python script into an interactive web app in minutes. We include it so students can present live, interactive analysis to any stakeholder without waiting for a developer.

Why We End With Business Analytics and ML

A senior analyst isn't just someone who cleans and visualizes data. They translate data into decisions. What does this funnel drop-off mean for revenue? Should we retain this customer segment? Which model predicts churn best? The final modules give students the language of business and a practical taste of machine

learning — so they can grow from analyst to decision-maker.

The 2-Month Structure

We designed this roadmap to be completed in exactly 8 weeks — 5 days a week, with clear weekly targets. Each module builds on the previous one. By the end of Week 8, a student will have touched every tool on a data analyst job description — and will have built real projects to prove it.

Course Overview — 10 Modules in 8 Weeks

#	Module	Weeks	Key Skills
1	Statistics & Probability	Week 1	Mean, SD, Hypothesis Testing, A/B Testing
2	Excel for Analysts	Week 1–2	Formulas, Pivot Tables, Dashboards, XLOOKUP
3	SQL — From Basics to Advanced	Week 2–3	SELECT → CTEs → Window Functions
4	Python for Data Analysis	Week 3–4	Pandas, NumPy, Matplotlib, Seaborn
5	Data Cleaning	Week 4	Missing values, Outliers, Transformation
6	Exploratory Data Analysis	Week 5	Univariate, Bivariate, Pattern Detection
7	Data Visualization + Streamlit	Week 5–6	Charts, Storytelling, Streamlit Apps
8	Power BI / Tableau	Week 6–7	Power Query, DAX, KPIs, Dashboards
9	Business Analytics	Week 7	KPIs, Funnels, Retention, Communication
10	Basic Machine Learning	Week 8	Regression, Trees, Model Evaluation

8-Week Day-by-Day Schedule

Week 1	
Mon	Statistics: Mean, Median, Mode, Variance, SD
Tue	Statistics: Probability, Normal Distribution
Wed	Statistics: Correlation, Covariance
Thu	Statistics: Hypothesis Testing, p-value
Fri	Statistics: Confidence Intervals, A/B Testing + Mini Quiz

Week 2	
Mon	Excel: Interface, Formulas — IF, COUNTIF, SUMIF
Tue	Excel: VLOOKUP, XLOOKUP, INDEX-MATCH
Wed	Excel: Pivot Tables — build from scratch
Thu	Excel: Charts, Conditional Formatting
Fri	Excel: Dashboard Project — end-to-end

Week 3	
Mon	SQL: SELECT, WHERE, ORDER BY, LIMIT
Tue	SQL: GROUP BY, HAVING, Aggregations
Wed	SQL: JOINS — INNER, LEFT, RIGHT, FULL
Thu	SQL: Subqueries and CTEs
Fri	SQL: Window Functions + Practice Set

Week 4	
Mon	Python: Variables, Loops, Functions, Data Types
Tue	Python: Pandas — DataFrames, filtering, groupby
Wed	Python: NumPy + Matplotlib basics
Thu	Python: Seaborn + EDA intro
Fri	Data Cleaning: Missing Values — MCAR, MAR, MNAR

Week 5	
Mon	Data Cleaning: Outlier Detection (IQR, Z-score)
Tue	Data Cleaning: Duplicates, Transformation, Feature Eng.
Wed	EDA: Univariate & Bivariate Analysis
Thu	EDA: Multivariate, Correlation Heatmaps
Fri	EDA: Story-Driven EDA Project

Week 6	
Mon	Visualization: Bar, Histogram, Scatter, Box Plots

Tue	Visualization: Heatmaps + Storytelling with Data
Wed	Streamlit: Building your first data app
Thu	Streamlit: Charts, filters, KPI cards in app
Fri	Streamlit: Deploy a mini analytics dashboard

Week 7

Mon	Power BI: Power Query & Data Modeling
Tue	Power BI: DAX — calculated columns vs measures
Wed	Power BI: KPIs, Slicers, Drill-through
Thu	Business Analytics: KPIs, Revenue, Retention metrics
Fri	Business Analytics: Funnel Analysis + Stakeholder Comm.

Week 8

Mon	ML: Linear Regression — theory + sklearn
Tue	ML: Logistic Regression + classification
Wed	ML: Decision Trees & Random Forest
Thu	ML: Model Evaluation — Accuracy, RMSE, ROC-AUC
Fri	CAPSTONE: End-to-end project presentation

1 Module 1: Statistics & Probability

Week 1

Why This Module?

Numbers without statistics are just noise. Statistics teaches analysts HOW to trust their findings. Without it, you might see a 5% sales increase and call it a trend — when it's just random variation. We teach this first so every later analysis is built on solid ground.

Topics Covered

• Mean, Median, Mode	• Hypothesis Testing
• Variance & Standard Deviation	• p-value & Significance
• Probability Basics	• Confidence Intervals
• Normal Distribution & Bell Curve	• A/B Testing
• Correlation & Covariance	• Type I & Type II Errors

Mini Project / Practice Task

- Run a real A/B test on sample marketing data — decide which campaign won and why.
- Calculate summary statistics on a sales dataset using Python.

2 Module 2: Excel for Analysts

Weeks 1–2

Why This Module?

Excel is still the most-used tool at 80%+ of companies. Before learning any software, an analyst must be fluent here. Managers open Excel, not Python scripts. Mastering Excel means you can contribute to a team from Day 1 of a job — no setup, no libraries, just open and go.

Topics Covered

• IF, COUNTIF, SUMIF formulas	• Pivot Charts
• VLOOKUP & XLOOKUP	• Slicers & Filters
• INDEX-MATCH	• Conditional Formatting
• Nested Formulas	• Dashboard Design in Excel
• Pivot Tables	• Data Validation

Mini Project / Practice Task

- Build a Sales Performance Dashboard with Pivot Tables, slicers, and KPI cards.

3 Module 3: SQL — From Basics to Advanced

Weeks 2–3

Why This Module?

Every company stores its data in a database. SQL is the ONLY way to get it out. No SQL = no data. It's listed as a requirement in 95% of data analyst job postings. We teach SQL end-to-end — from a simple SELECT to complex Window Functions — because real-world queries are rarely simple.

Topics Covered

• SELECT, FROM, WHERE	• FULL OUTER JOIN & SELF JOIN
• ORDER BY, LIMIT, DISTINCT	• Subqueries
• GROUP BY & HAVING	• CTEs (WITH clause)
• Aggregate Functions (SUM, COUNT, AVG)	• Window Functions (RANK, ROW_NUMBER, LAG)
• INNER JOIN / LEFT JOIN / RIGHT JOIN	• Performance & Indexing basics

Mini Project / Practice Task

- Analyze an e-commerce database: top products by revenue, monthly trends, customer repeat rate.
- Build a cohort retention query using Window Functions.

4 Module 4: Python for Data Analysis

Weeks 3–4

Why This Module?

Python automates everything Excel can't scale. Pandas can process millions of rows in seconds. Matplotlib and Seaborn produce publication-quality charts. We teach Python not as a programming course but as a data tool — every concept is taught through a data problem, not an abstract exercise.

Topics Covered

• Variables, Data Types, Lists, Dicts	• NumPy: arrays, operations
• Loops & Functions	• Matplotlib: line, bar, scatter
• Pandas: DataFrames & Series	• Seaborn: distribution, heatmap, pairplot
• Pandas: read_csv, filtering, groupby	• Reading & Writing files (CSV, Excel)
• Pandas: merge, concat, pivot_table	• Lambda functions & apply()

Mini Project / Practice Task

- Load a superstore dataset, clean it with Pandas, and produce 5 insights with Seaborn visualizations.

5 Module 5: Data Cleaning

Week 4

Why This Module?

60–80% of a data analyst's job is cleaning raw, messy data before any analysis can begin. Real-world data has nulls, outliers, duplicates, wrong formats, and inconsistent labels. If you can't clean it, you can't trust your analysis. This module treats data cleaning as a core skill, not an afterthought.

Topics Covered

• Missing Values: MCAR, MAR, MNAR	• Data Type Conversion
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• Imputation strategies (mean, median, mode, forward fill)	• String Cleaning (strip, replace, regex)
• Outlier Detection: IQR method	• Date Parsing & Formatting
• Outlier Detection: Z-score method	• Data Transformation (normalization, scaling)
• Duplicate Identification & Removal	• Feature Engineering basics

Mini Project / Practice Task

→ Given a 'dirty' HR or sales dataset, apply full cleaning pipeline and document every decision.

6**Module 6: Exploratory Data Analysis (EDA)**

Week 5

Why This Module?

EDA is how analysts discover the story inside data before building anything. It's the detective phase — looking for patterns, anomalies, trends, and relationships. Without EDA, you might build a dashboard on wrong assumptions. We teach EDA as a structured process: from single-variable summaries to multi-variable relationships.

Topics Covered

• Univariate Analysis (describe, histograms)	• Correlation Matrix
• Bivariate Analysis (scatter, cross-tab)	• Outlier Visualization (box plots)
• Multivariate Analysis (pairplot, heatmap)	• Time Series EDA
• Distribution Analysis	• Pattern Detection
• Skewness & Kurtosis	• Story-Driven EDA Narrative

Mini Project / Practice Task

→ Perform full EDA on a retail or healthcare dataset and present a 5-insight story with charts.

7**Module 7: Data Visualization + Streamlit**

Weeks 5–6

Why This Module?

An insight that can't be communicated clearly is worthless. We teach static visualizations (Matplotlib, Seaborn) for reports and presentations, but we also include Streamlit — a Python library that lets you turn any script into a live, interactive data app in minutes. This is increasingly expected in modern analyst roles and sets students apart in interviews.

Topics Covered

• Bar Charts & Grouped Bar Charts	• Heatmaps (correlation & matrix)
• Line Charts (trends over time)	• Storytelling with Data principles
• Scatter Plots (relationships)	• Choosing the right chart for the data
• Histograms (distributions)	• Streamlit: layout, widgets, charts
• Box Plots (spread & outliers)	• Streamlit: KPI cards, filters, deploy

Mini Project / Practice Task

→ Build a Streamlit app with filters, KPI metrics, and 3 interactive charts from a sales dataset.

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Module 8: Power BI & Tableau Dashboards

Weeks 6–7

Why This Module?

Power BI is the #1 BI tool in corporate India and worldwide for business dashboards. It takes data analysis to management-level presentation. We teach Power BI end-to-end — from importing data to building KPI dashboards with DAX measures. Tableau is introduced for completeness and global market awareness.

Topics Covered

• Power Query: import, transform, merge	• Slicers, Drill-through, Bookmarks
• Data Modeling: relationships, star schema	• KPI Cards & Gauge Charts
• DAX: calculated columns vs measures	• Report Publishing & Sharing
• DAX: SUM, CALCULATE, FILTER	• Tableau: basic intro (charts, filters)
• Time Intelligence (YTD, MTD)	• Dashboard design best practices

Mini Project / Practice Task

→ Build a Sales Analytics Power BI report with 3 pages: Overview, Product Performance, Customer Analysis.

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Module 9: Business Analytics

Week 7

Why This Module?

Tools are only half the job. The other half is understanding what the business is asking. What does a 20% churn rate mean for revenue? Is a 3% conversion rate good or bad? This module teaches analysts to speak the language of business — KPIs, funnels, retention, and cohort analysis — and to present findings that lead to action.

Topics Covered

• What is a KPI? How to define good KPIs	• Funnel Analysis (Awareness → Purchase)
• Revenue Metrics (ARR, MRR, ARPU)	• RFM Segmentation
• Customer Lifetime Value (CLV)	• Net Promoter Score (NPS)
• Churn Analysis	• Stakeholder Communication
• Retention Curves & Cohort Analysis	• Presenting Data to Non-Technical Audiences

Mini Project / Practice Task

→ Analyze a SaaS company dataset: calculate churn, CLV, and build a retention funnel.

→ Present findings as a 5-slide business summary (no jargon).

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Module 10: Basic Machine Learning

Week 8

Why This Module?

A growing number of analyst roles now expect basic ML knowledge — not to build production models, but to understand, evaluate, and communicate model outputs. This module gives students a practical foundation in supervised learning so they can collaborate with data scientists and add predictive analysis to their toolkit.

Topics Covered

• What is ML? Supervised vs Unsupervised	• Random Forest
• Train/Test Split	• Model Evaluation: Accuracy, Precision, Recall
• Linear Regression (predict a number)	• RMSE, MAE for regression
• Logistic Regression (predict yes/no)	• ROC-AUC Curve
• Decision Trees	• Overfitting & Cross-Validation

Mini Project / Practice Task

→ Build a churn prediction model (Logistic Regression + Random Forest) on a telecom dataset.

→ Evaluate and compare both models. Write a 1-page business summary of results.

You're Ready.

By the end of these 8 weeks, your students will have covered every skill listed on a data analyst job description — and built real projects to prove it. They won't just know the tools. They'll know why each tool exists, when to use it, and how to explain their analysis to any audience.
